DM872

Math Optimization at Work

Lagrangian Relaxation

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[*Partly based on slides by David Pisinger, DIKU (now DTU)*]

Outline

1. Relaxations and Bounds

2. Subgradient Optimization

3. LR in IP

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Relaxation

In branch and bound we find upper bounds by relaxing the problem

Definition (Relaxation)

$$\max_{\boldsymbol{x} \in P} g(\boldsymbol{x}) \geq \left\{ \begin{array}{l} \max_{\boldsymbol{x} \in P} f(\boldsymbol{x}) \\ \max_{\boldsymbol{x} \in S} g(\boldsymbol{x}) \end{array} \right\} \geq \max_{\boldsymbol{x} \in S} f(\boldsymbol{x})$$

- P: candidate solutions
- $S \subseteq P$ feasible solutions
- $g(\boldsymbol{x}) \geq f(\boldsymbol{x})$

Which constraints should be relaxed?

- Quality of bound (tightness of relaxation)
- Remaining problem can be solved efficiently
- Proper multipliers can be found efficiently
- Constraints difficult to formulate mathematically
- Constraints which are too expensive to write up

Relevant Relaxations

Different relaxations

- LP-relaxation
- Deleting constraint
- Lagrange relaxation
- Surrogate relaxation
- Semidefinite relaxation

Relaxations are often used in combination.

Tighter

Best surrogate relaxation

Best Lagrangian relaxation

LP relaxation

Surrogate Relaxation

Integer Programming Problem: $\max\{cx \mid Ax \leq b, Dx \leq e, x \in \mathbb{Z}_+^n\}$
Relax complicating constraints $Dx \leq e$.
Surrogate Relax $Dx \leq e$ using multipliers $\lambda \geq 0$, i.e., add together constraints using weights λ

$$\begin{aligned} z_{SR}(\lambda) = \max\ & cx \\ \text{s.t. } & Ax \leq b \\ & \lambda Dx \leq \lambda e \\ & x \in \mathbb{Z}_+^n \end{aligned}$$

Proposition

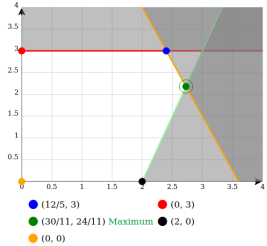
Optimal Solution to the surrogate relaxed problem gives an upper bound on original problem

<u>Proof:</u> by showing that it is a relaxation: $Dx \leq e \implies \lambda Dx \leq \lambda e, \lambda \geq 0$ □

Each multiplier λ_i is a weighting of the corresponding constraint
If λ_i large $\implies$ constraint satisfied (at expenses of other constraints)
If $\lambda_i = 0 \implies$ drop the constraint

Surrogate Relaxation: Example

$$\begin{array}{rrl} \text{maximize} & 4x_1 + & x_2 \\ \text{subject to} & 3x_1 - & x_2 \leq 6 \\ & & x_2 \leq 3 \\ & 5x_1 + & 2x_2 \leq 18 \\ & x_1, & x_2 \geq 0, \text{integer} \end{array}$$



IP solution $(x_1, x_2) = (2, 3)$ with $z_{IP} = 11$
LP solution $(x_1, x_2) = (\frac{30}{11}, \frac{24}{11})$ with $z_{LP} = \frac{144}{11} = 13.1$
First and third constraints complicating, surrogate relax using multipliers $\lambda_1 = 2$ and $\lambda_3 = 1$:

$$\begin{array}{rrl} \text{maximize} & 4x_1 + & x_2 \\ \text{subject to} & & x_2 \leq 3 \\ & 11x_1 & \leq 30 \\ & x_1, & x_2 \geq 0, \text{integer} \end{array}$$

Solution $(x_1, x_2) = (2, 3)$ with $z_{SR} = 4 \cdot 2 + 3 = 11$. Upper bound.

Lagrangian Relaxation

Integer Linear Programming problem

$$\begin{aligned} z = \max\ & cx \\ \text{s.t. } & Ax \leq b \\ & Dx \leq e \\ & x \in \mathbb{Z}^n_+ \end{aligned}$$

We relax the $Dx \leq e$ constraints:

Lagrangian Relaxation, $\lambda \geq 0$:

$$\begin{aligned} z_{LR}(\lambda) = \max\ & cx - \lambda(Dx - e) \\ \text{s.t. } & Ax \leq b \\ & x \in \mathbb{Z}^n_+ \end{aligned}$$

optimizes over the x variables with λ fixed

Lagrangian Dual Problem

$$z_{LD} = \min_{\lambda \geq 0} z_{LR}(\lambda)$$

optimizes over the λ variables with x fixed

Tightness of Relaxations (1/2)

Integer Linear Programming problem

$$\begin{aligned} z = \max\ & cx \\ \text{s.t. } & Ax \leq b \\ & Dx \leq e \\ & x \in \mathbb{Z}^n_+ \end{aligned}$$

It corresponds to:

$$\boxed{z = \max \left\{ cx \ : \ x \in \operatorname{conv}(Ax \leq b, Dx \leq e, x \in \mathbb{Z}^n_+) \right\}}$$

LP-relaxation:

$$\boxed{z_{LP} = \max \left\{ cx \ : \ x \in Ax \leq b, Dx \leq e, x \in \mathbb{R}^n_+ \right\}}$$

Lagrangian Relaxation, $\lambda \geq 0$:

$$\begin{aligned} z_{LR}(\lambda) = \max\ & cx - \lambda(Dx - e) \\ \text{s.t. } & Ax \leq b \\ & x \in \mathbb{Z}^n_+ \end{aligned}$$

Lagrange Dual Problem

$$z_{LD} = \min_{\lambda \geq 0} z_{LR}(\lambda)$$

with best multipliers λ it corresponds to:

$$\boxed{z_{LD} = \max \left\{ cx \ : \ Dx \leq e, x \in \operatorname{conv}(Ax \leq b, x \in \mathbb{Z}^n_+) \right\}}$$

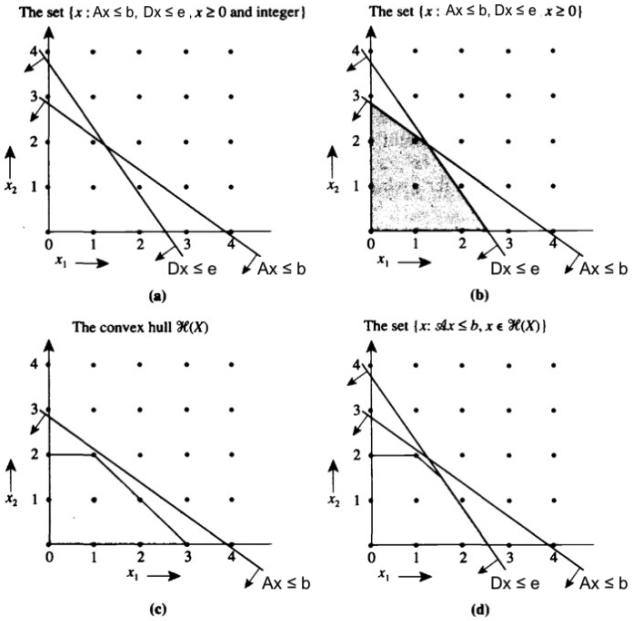


Fig 16.6 from [AMO]

Tightness of Relaxations (2/2)

Surrogate Relaxation, $\lambda \geq 0$

$$\begin{aligned} z_{SR}(\lambda) = \max\ & cx \\ \text{s.t. } & Ax \leq b \\ & \lambda Dx \leq \lambda e \\ & x \in \mathbb{Z}^n_+ \end{aligned}$$

Surrogate Dual Problem

$$z_{SD} = \min_{\lambda \geq 0} z_{SR}(\lambda)$$

with best multipliers λ:

$$\boxed{z_{SD} = \max \left\{ cx \ : \ x \in \operatorname{conv}(Ax \leq b, \lambda Dx \leq \lambda e, x \in \mathbb{Z}^n_+) \right\}}$$

$\rightsquigarrow$ Best surrogate relaxation (i.e., best λ multipliers) is tighter than best Lagrangian relaxation.

Tightness of Relaxations

We have proved that:

Proposition (Tightness of Relaxations)

$$z_{SD} \leq z_{LD} \leq z_{LP}$$

Hence, the best dual bound would be given by z_{SD} but we do not have a good method to determine the λ

Relaxation strategies

Which constraints should be relaxed

- "the complicating ones"
- remaining problem is polynomially solvable
  (e.g. min spanning tree, assignment problem, linear programming)
- remaining problem is totally unimodular
  (e.g. network problems)
- remaining problem is NP-hard but good techniques exist
  (e.g. knapsack)
- constraints which cannot be expressed in MIP terms
  (e.g. cutting)
- constraints which are too extensive to express
  (e.g. subtour elimination in TSP)

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Subgradient Optimization of Lagrangian Multipliers

$$\begin{aligned} z = \max\ & cx \\ \text{s.t.}\ & Ax \leq b \\ & Dx \leq e \\ & x \in \mathbb{Z}^n_+ \end{aligned}$$

Lagrange Relaxation, multipliers $\lambda \geq 0$

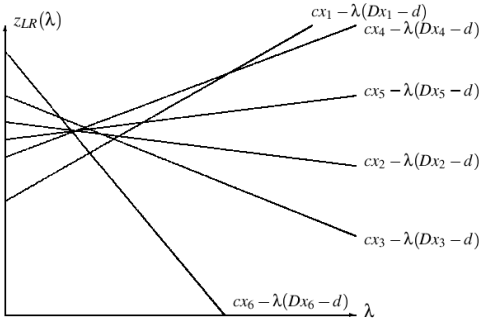
$$\begin{aligned} z_{LR}(\lambda) = \max\ & cx - \lambda(Dx - e) \\ \text{s.t.}\ & Ax \leq b \\ & x \in \mathbb{Z}^n_+ \end{aligned}$$

Lagrange Dual Problem

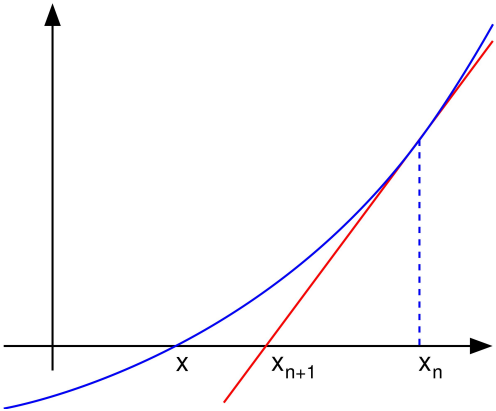
$$z_{LD} = \min_{\lambda \geq 0} z_{LR}(\lambda)$$

- We do not need best multipliers in B&B algorithm
- Subgradient optimization fast method
- Works well due to convexity
- Roots in nonlinear programming, Held and Karp (1971)

Subgradient optimization, motivation



Lagrange function $z_{LR}(\lambda)$ is piecewise linear and convex



Netwon-like method to minimize a function in one variable

Digression: Gradient methods

Gradient methods are iterative methods:

- find a descent direction with respect to the objective function f
- move x in that direction by a step size

The descent direction can be computed by various methods, such as gradient descent, Newton method and others. The step size can be computed either exactly or loosely by solving a line search problem.

Gradient descent algorithm:

Set iteration counter $k = 0$, and make an initial guess x_0 for the minimum
Repeat:
  Compute a descent direction $d_k = -\nabla(f(x_k)$
  Choose α_k to minimize $f(x_k + \alpha d_k)$ over $\alpha_k \in \mathbb{R}_+$
  Update $x_{k+1} \leftarrow x_k + \alpha_k d_k$, and $k \leftarrow k + 1$
Until $\|\nabla f(x_k)\| < tolerance$

We will set α_k 'loosely' by taking small enough values $\alpha_k > 0$

Newton-Raphson method and Newton method

Another example of gradient descent algorithm:

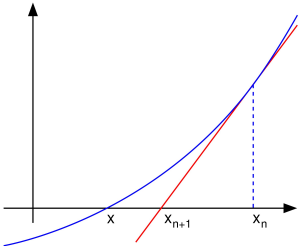
Find zeros of a real-valued, derivable function

$$x : f(x) = 0\,.$$

- Start with a guess x_0
- Repeat:
  Move to a better approximation

$$x_{k+1} = x_k - \frac{f(x_k)}{f'(x_k)}$$

until a sufficiently accurate value is reached.



Geometrically, $(x_{k+1}, 0)$ is the intersection with the x-axis of a line tangent to f at $(x_k, f(x_k))$.

$$f'(x_k) = \frac{\Delta y}{\Delta x} = \frac{f(x_k) - 0}{x_k - x_{k+1}}.$$

Subgradient

Subgradient: Generalization of gradients to non-differentiable, continuous functions.

Definition

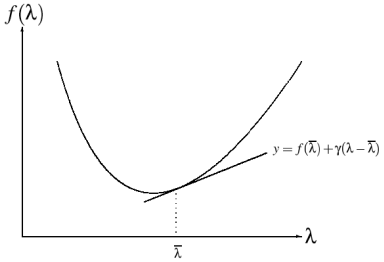
An m-vector γ is subgradient of $f(\lambda)$ at $\bar{\lambda}$ if

$$f(\lambda) \geq f(\bar{\lambda}) + \gamma(\lambda - \bar{\lambda})$$

The inequality says that the hyperplane

$$y = f(\bar{\lambda}) + \gamma(\lambda - \bar{\lambda})$$

is tangent to $y = f(\lambda)$ at $\lambda = \bar{\lambda}$ and supports the epigraph of f at $(\lambda, f(\lambda))$ from below



Proposition

Given a choice of nonnegative multipliers $\bar{\lambda}$, if x' is an optimal solution to $z_{LR}(\bar{\lambda})$ then

$$\gamma = e - Dx'$$

is a subgradient of $z_{LR}(\lambda)$ at $\lambda = \bar{\lambda}$.

<u>Proof:</u> Note that for us in the LD problem: $f(\lambda) = \max_{Ax \leq b} (cx - \lambda(Dx - e))$.
We wish to prove that the inequality from the subgradient definition holds:

$$\max_{Ax \leq b} (cx - \lambda(Dx - e)) \geq \max_{Ax \leq b} \left(cx - \bar{\lambda}(Dx - e)\right) + \gamma(\lambda - \bar{\lambda})$$

Indeed:

- We note that in the RHS: $\max_{Ax \leq b} \left(cx - \bar{\lambda}(Dx - e)\right) = (cx' - \bar{\lambda}(Dx' - e))$ because x' is by hypothesis the optimal solution of $f(\bar{\lambda})$.
- Rewriting the inequality using the hypothesis on γ we have:
$$\max_{Ax \leq b} (cx - \lambda(Dx - e)) \;\geq\; (cx' - \bar{\lambda}(Dx' - e)) + (e - Dx')(\lambda - \bar{\lambda}) \;=\; cx' - \lambda(Dx' - e)$$
The right most part is the evaluation of the left most problem at a single feasible solution. Hence, it can be at most $\leq$ of the left most part, as we wanted to prove.

Intuition

Lagrange dual:

$$\begin{aligned} &\min z_{LR}(\lambda) = cx - \lambda(Dx - e) \\ &\text{s.t. } Ax \leq b \\ &\quad\;\; x \in \mathbb{Z}^n_+ \end{aligned}$$

Gradient in x' is

$$\gamma = e - Dx'$$

Subgradient Iteration

Recursion

$$\lambda_{k+1} = \max\{\lambda_k - \theta\gamma_k, 0\}$$

where $\theta_k > 0$ is step-size

If $\gamma_k > 0$ and θ_k is sufficiently small $z_{LR}(\lambda)$ will decrease.

- Small θ_k slow convergence
- Large θ_k unstable

Held and Karp procedure (gradient descent)

Initially

$$\lambda^0 = [0, \ldots, 0]$$

compute the new multipliers by recursion

$$\lambda_{i,k+1} := \begin{cases} \lambda_{i,k} & \text{if } |\gamma_i| \leq \epsilon \\ \max(\lambda_{i,k} - \theta_k \gamma_i, 0) & \text{if } |\gamma_i| > \epsilon \end{cases}$$

where γ_i is the ith component of the subgradient.
The step size θ_k is defined by

$$\theta_k = \mu \frac{z_{LR}(\lambda_k) - \underline{z}}{\sum_i \gamma_i^2}$$

where μ is an appropriate constant and $\underline{z}$ a heuristic primal (lower) bound for the orginal ILP problem.
E.g. $\mu = 1$ and halved if upper bound not decreased in 20 iterations.

Lagrangian Relaxation and LP

For an LP-problem where we Lagrange relax all constraints

- Dual variables are best choice of Lagrange multipliers
- Lagrange relaxation and LP "relaxation" give same bound

Gives a clue to solve LP-problems without Simplex

- Iterative algorithms
- Polynomial algorithms

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Lagrangian Relaxation in Integer Programming

Original Problem (OP)

$$\begin{aligned} z = \max\ & \boldsymbol{c}^T \boldsymbol{x} \\ \text{s.t. } & A\boldsymbol{x} \leq \boldsymbol{b} \\ & D\boldsymbol{x} \leq \boldsymbol{e} \\ & \boldsymbol{x} \geq 0 \\ & \boldsymbol{x} \text{ integer} \end{aligned}$$

Lagrangian Relaxation Problem (LR) $\boldsymbol{\lambda} \geq 0$:

$$\begin{aligned} z_{LR}(\boldsymbol{\lambda}) = \max\ & \boldsymbol{c}^T \boldsymbol{x} - \boldsymbol{\lambda}^T (D\boldsymbol{x} - \boldsymbol{e}) \\ \text{s.t. } & A\boldsymbol{x} \leq \boldsymbol{b} \\ & \boldsymbol{x} \geq \boldsymbol{0} \\ & \boldsymbol{x} \text{ integer} \end{aligned}$$

- Note that in Lagrangian Relaxation the integrality constraint is not relaxed
- z_{LP} objective function value of linear relaxation of OP
- $z_{LD} = \min_{\boldsymbol{\lambda} \geq 0} z_{LR}(\boldsymbol{\lambda})$ Lagrangian dual problem.

Facts

$z_{LP} \geq z$	because relaxation
$z_{LR} \geq z$	because relaxation
$z_{LR} \geq z_{LD}$	because of definition
$z_{LP} \geq z_{LD}$	this is not trivial (see next slide) but important for motivating the use of Lagrangian Relaxation in Integer Programming

- Motivation A: if $z_{LP} > z_{LD}$ then LR gives us a better bound to in B&B.
- Motivation B: if $z_{LP} = z_{LD}$ LR can still be worth because Z_{LD} can be found more easily than with LP
- Motivation C: in any case LR gives us an alternative way to solve the problem. It is an heuristics way with the rare chance of getting also a dual bound and eventualy a provable optimal solution.

For a maximization problem: $z \leq z_{LD} \leq \left(z_{LR} \lesseqgtr z_{LP}\right)$

Proposition

$$z_{LP} \geq z_{LD}$$

<u>Proof:</u> There are two ways of prooving this:

1. via the convexification argument as in the previous slides (see also sec 16.4 of [AMO])
2. via the duality argument also presented in sec 8 of [Fi]

Let's use the second.

$$\begin{aligned} z_{LD} &= \min_{\lambda \geq 0} z_{LR}(\lambda) = \\ &= \min_{\lambda \geq 0} \left\{ \max_x \left\{ c^T x - \lambda^T (Dx - e) \mid Ax \leq b, x \geq 0, x \text{ integer} \right\} \right\} \leq \\ &\leq \min_{\lambda \geq 0} \left\{ \max_x \left\{ c^T x - \lambda^T (Dx - e) \mid Ax \leq b, x \geq 0 \right\} \right\} = \end{aligned}$$

$$= \min_{\lambda \geq 0} \left\{ \underbrace{\max_{x} \left\{ c^T x - \lambda^T (Dx - e) \mid Ax \leq b, x \geq 0 \right\}} \right\} =$$

$$\begin{aligned} &\max\ cx - \lambda(Dx - e) \\ &\mu :\ Ax \leq b \\ &\quad\ \boldsymbol{x} \geq 0 \end{aligned} \quad \overset{Dual}{\Longrightarrow} \quad \begin{aligned} &\min\ \mu^T b + \lambda^T e \\ &\quad \mu^T A + \lambda^T D \geq c \\ &\quad \mu \geq 0 \\ &\quad \lambda \geq 0 \end{aligned}$$

$$= \min_{\lambda \geq 0} \left\{ \min_{\mu \geq 0} \left\{ \mu^T b + \lambda^T e \mid \mu^T A + \lambda^T D \geq c, \mu \geq 0 \right\} \right\} =$$

$$= \underbrace{\min_{\lambda \geq 0} \left\{ \min_{\mu \geq 0} \left\{ \mu^T b + \lambda^T e \mid \mu^T A + \lambda^T D \geq c, \mu \geq 0 \right\} \right\}} =$$

$$\begin{aligned} \min\ & \mu^T b + \lambda^T e \\ x :\ & \mu^T A + \lambda^T D \geq c \\ & \mu \geq 0 \\ & \lambda \geq 0 \end{aligned} \quad \xrightarrow{Dual} \quad \begin{aligned} \max\ & c^T x \\ & Ax \leq b \\ & Dx \leq e \\ & x \geq 0 \end{aligned}$$

$$= z_{LP} \quad \square$$

Corollary

$z_{LD} = z_{LP}$ when the LR problem has the integrality property

Proof: The only inequality introduced in the derivations of the previous proof becomes equality as well. □